* Notation: Same as §3.1 - 3.3 in extended delta notes.

Theorem 3.2.1 (by Schiffmann and Vasserot).

Let $\Gamma(\omega,y)$ be a rational function s.t.

$$\frac{\Gamma(\omega,y)}{\Gamma(y,\omega)} = \Omega\left[-\frac{\omega}{y}\hat{\pi}\right] = \frac{\left(1-qt\frac{\omega}{y}\right)\left(1-\frac{1}{q}\cdot\frac{\omega}{y}\right)\left(1-\frac{1}{t}\frac{\omega}{y}\right)}{\left(1-\frac{1}{qt}\cdot\frac{\omega}{y}\right)\left(1-q\cdot\frac{\omega}{y}\right)\left(1-t\cdot\frac{\omega}{y}\right)} \quad \text{———} \quad (*)$$

Then there is an algebra isomorphism

$$\psi : S_\Gamma \longrightarrow \mathcal{E}^+ \text{ given by}$$
$$z_i^a \longmapsto p_i[-Mx^{i,a}].$$

If $\Gamma(\omega,y)$ and $\Gamma'(\omega,y)$ both satisfy $(*)$, then $\frac{\Gamma'(\omega,y)}{\Gamma(\omega,y)} = \frac{\Gamma'(y,\omega)}{\Gamma(y,\omega)}$. Hence $h(\omega,y) := \frac{\Gamma'(\omega,y)}{\Gamma(\omega,y)}$ satisfies $h(\omega,y) = h(y,\omega)$.

By Lemma 3.1.1, $I_{\Gamma'} = I_\Gamma$ and $S_{\Gamma'} = S_\Gamma$. Hence for any $\Gamma(\omega,y)$ satisfying $(*)$, we can fix

$$S = S_\Gamma,\ I = I_\Gamma,\ S^\ell = S_\Gamma^\ell,\ I^\ell = I_\Gamma^\ell.$$

Note that even though S is independent of the choice of Γ, the concrete shuffle algebra does.

The following two choices of Γ satisfying $(*)$ turn out to be convenient:

$$\hat{\Gamma}(\omega,y) = \frac{1-qt\frac{\omega}{y}}{\left(1-\frac{y}{\omega}\right)\left(1-q\frac{\omega}{y}\right)\left(1-t\frac{\omega}{y}\right)}$$

$$\check{\Gamma}(\omega,y) = \left(1-\frac{\omega}{y}\right)\left(1-q\frac{y}{\omega}\right)\left(1-t\frac{y}{\omega}\right)\left(1-qt\frac{\omega}{y}\right)$$

$$\left(\text{verify: } \frac{\hat{\Gamma}(\omega,y)}{\hat{\Gamma}(y,\omega)} = \frac{\left(1-qt\frac{\omega}{y}\right)}{\left(1-q\frac{\omega}{y}\right)\left(1-t\frac{\omega}{y}\right)} \cdot \frac{\left(1-\frac{\omega}{y}\right)\left(1-q\frac{y}{\omega}\right)\left(1-t\frac{y}{\omega}\right)}{\left(1-\frac{y}{\omega}\right)\left(1-qt\frac{y}{\omega}\right)} = \frac{\left(1-qt\frac{\omega}{y}\right)}{\left(1-q\frac{\omega}{y}\right)\left(1-t\frac{\omega}{y}\right)} \cdot \overbrace{\frac{1}{y}\cdot\cancel{\frac{q}{\omega}\cdot\frac{y}{\omega}\cdot t\frac{y}{\omega}(y-\omega)}}^{qt\cdot\frac{y}{\omega^2}(y-\omega)} \cdot \frac{1}{\omega}\cdot\underbrace{\cancel{\left(-qt\frac{y}{\omega}\right)(\omega-y)}}_{qt\frac{y}{\omega^2}\cdot(y-\omega)} \left(1-\frac{1}{qt}\cdot\frac{\omega}{y}\right) = \Omega\left[-\frac{\omega}{y}\hat{\pi}\right]\right)$$

$$\left(\check{\Gamma}(\omega,y) = \hat{\Gamma}(\omega,y)\cdot\underbrace{\left(1-\frac{y}{\omega}\right)\left(1-q\frac{y}{\omega}\right)\left(1-q\frac{\omega}{y}\right)\left(1-t\frac{y}{\omega}\right)\left(1-t\frac{\omega}{y}\right)}_{\text{symmetric in } \omega,y}\right)$$

For either of these choices of Γ, define $\psi_\Gamma := \psi\circ\sigma_\Gamma^{-1} : S_\Gamma \xrightarrow{\cong} \mathcal{E}^+$, so we have a commutative diagram with all arrows isomorphisms:

$$\begin{tikzcd}
S_{\hat{\Gamma}} \arrow[dr, "\psi_{\hat{\Gamma}}"] & \\
S \arrow[u, "\sigma_{\hat{\Gamma}}"] \arrow[r, "\psi"] \arrow[d, "\sigma_{\check{\Gamma}}"] & \mathcal{E}^+ \\
S_{\check{\Gamma}} \arrow[ur, "\psi_{\check{\Gamma}}"'] &
\end{tikzcd}$$